

- 6 (a) Show that the curve with Cartesian equation

$$(x^2 + y^2)^2 = 36(x^2 - y^2)$$

has polar equation $r^2 = 36 \cos 2\theta$. [3]

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The curve C has polar equation $r^2 = 36 \cos 2\theta$, for $-\frac{1}{4}\pi \leq \theta \leq \frac{1}{4}\pi$.

- (b) Sketch C and state the maximum distance of a point on C from the pole. [3]

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- (c) Find the area of the region enclosed by C . [2]

- (d) Find the maximum distance of a point on C from the initial line, giving the answer in exact form. [6]

[illegible]